

ClimateFeel, a Sensory Experience of Climate Change

Yidong Huang

LIG Laboratory

Université Grenoble Alpes, CNRS, Grenoble INP

Grenoble, France

yidong.huang@univ-grenoble-alpes.fr

Yann Laurillau

LIG Laboratory

Université Grenoble Alpes, CNRS, Grenoble INP

Grenoble, France

yann.laurillau@univ-grenoble-alpes.fr

Abstract

Direct personal experience shapes how people perceive climate change, yet opportunities for sensory engagement are limited, making bottom-up perception difficult. HCI approaches, data visualisation, storytelling, immersive systems, and eco-feedback, primarily rely on visual and textual modalities, leaving bodily sensation underexplored as a design resource. In this Interactive Demo paper, we present ClimateFeel, a system and interaction technique that maps historical temperature data to controlled thermal stimuli delivered at the fingertip. Users can directly feel differences in temperature over time through our system. Our contributions are twofold: (1) introducing a new interaction technique that allows sensory comparison of climate data, and (2) demonstrating bottom-up, sensory-driven interaction to support experiential understanding of climate change.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**.

Keywords

Thermal feedback, thermal haptics, climate change communication

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1 Introduction

Direct personal experience plays an important role in shaping how people perceive and make sense of climate change. Prior research shows that climate change perception emerges from the interplay between top-down, analytically driven processes (e.g., reasoning about statistics and models) and bottom-up, sensory-driven processes rooted in bodily experience [1, 4, 6, 7, 19, 22, 32, 33]. While both processes are important, there are many factors that disrupts climate change perception through direct personal experiences [1, 3, 5, 6, 10, 12–14, 18, 21, 32, 33, 35], making a bottom-up process driven climate change perception difficult.

In HCI, climate change communication has been explored through different approaches: data visualisation, data storytelling, immersive visualisation, eco-feedback systems and more [8, 9, 16, 20, 24, 25]. However, these approaches predominantly rely on visual and other modalities such as text and speech, offering limited support for a direct, embodied sensory experience. As a result, bottom-up perceptual processes that are grounded in bodily sensation remain underexplored as a design resource for climate change communication.

In this Interactivity paper, we present *ClimateFeel*, a novel interaction technique and demonstrable system that enables users to explore and understand climate data through touch-based thermal stimuli. The system maps different historical temperature records to controlled thermal stimuli delivered at the fingertip, allowing users to directly feel the changes in sensation as the result of increasing temperature due to climate change. Our contribution through this demonstration is twofold: (1) we introduce an interaction technique to allow sensory comparison of climate data; (2) we explore using bottom-up, sensory-driven interaction to support experiential understanding of gradual and heterogeneous climate change.

2 Background and related work

2.1 Perception of climate change

Research on climate change perception shows that people frequently rely on readily available cues when judging climate trends, with thermal sensation and local weather experiences playing a particularly influential role [1, 6, 7, 12, 21, 35]. Visceral experiences such as feeling unusually warm or cold have been shown to also shape beliefs and concerns about climate change [18]. However, direct perception of climate change is constrained in several ways. Judgements about long-term climatic change often depend on memory of past weather, which decays over time and is subject to distortion [5]. Pre-existing beliefs further bias recall and interpretation through motivated reasoning, leading to asymmetric perception of climate anomalies [12, 13, 22]. Structural factors also limit sensory exposure: indoor climate control technologies isolates individuals from environmental temperature variation [3], while the global and long-term nature of climate change exceeds the temporal and spatial scope of everyday experience of most individuals. Together, these factors restrict opportunities for bottom-up perception and reinforce reliance on abstract, top-down cognitive process. These limitations suggest an opportunity for interaction techniques that can reintroduce sensory experience into climate change perception and communication, enabling users to engage bodily with climate change.



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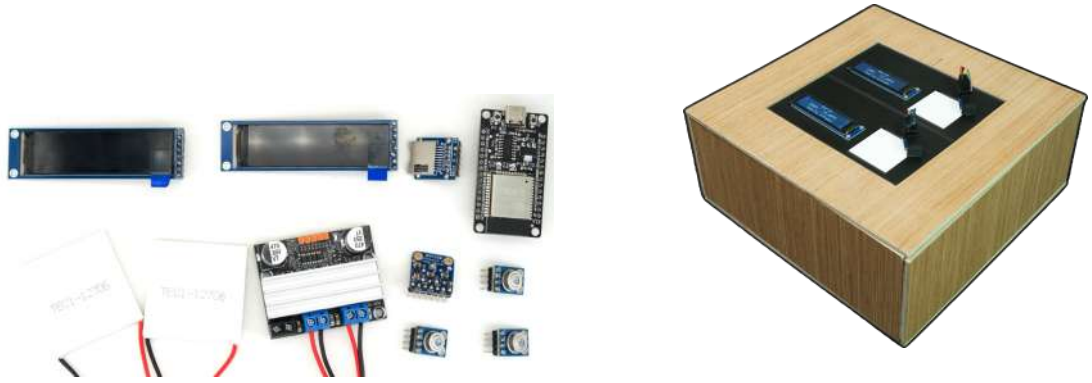


Figure 1: Right: The main components of *ClimateFeel*. Left: The assembled prototype in its enclosure.

2.2 HCI and Climate Change Communication

The HCI community has demonstrated strong engagement with climate change communication through a diverse range of approaches, prior work documents a growing number of HCI research projects focused on climate communication and many of these systems aim to connect climate information to personal experience through narratives, scenarios, or contextualised data [9, 20, 24, 25].

Despite this diversity, most existing approaches rely heavily on visual and symbolic representation, favouring the top-down process that is driven by analytical reasoning and reflection. We found few works on climate change communication through alternative sensory modalities, even immersive systems typically prioritise visual immersion (e.g. [16, 26, 30]). Consequently, climate change communication in HCI has largely focused on supporting the top-down cognitive process, with comparatively little attention to the embodied, sensory-driven, bottom up process. This gap motivates exploration of alternative sensory modalities that can support experiential engagement with climate data.

2.3 Thermal Sensation as an Interaction Modality

Prior work demonstrates the potential of using thermal stimulation in interaction design to convey information and evoke meaningful bodily experiences [34]. Thermal feedback has been used to communicate abstract information such as emotional response [11, 31], support decision-making [15], and enrich interaction through embodied sensation [17, 23, 27, 28].

In our prior work, the *Feel-Heat? Feel-It!* thermostat demonstrated that users can accurately perceive and compare thermostat setpoints through touch based thermal feedback [15]. This work highlights that temperature, an abstract numerical value, can be represented in a perceptually grounded manner by thermal stimuli.

Additionally, thermal sensation can be used to reinforce techniques such as climate analogues [2] often employed in climate science popularization, by conveying shifts in climate conditions, particularly temperature, through concrete, perceptible differences in sensory experience.

Building on this foundation, we extend this interaction technique from the domain of environmental control to the representation of long-term, large-scale climate data. In doing so, we explore how

thermal sensation can be used in interaction design to support bottom-up perceptual processes to complement exiting approaches in climate change communication.

3 ClimateFeel

We present *ClimateFeel*, a proof-of-concept interactive system that enables users to explore and compare climate temperature data through touch-based thermal sensation. The system is suitable for a hands-on interactive live demo experience.

3.1 Interaction Concept

Users select two climate data points (e.g. average surface temperature values from different years, seasons, or locations) via a graphical user interface (GUI). The system then translates the temperature values of selected data points into corresponding thermal stimuli presented simultaneously using two Peltier modules to allow easy comparison of the two thermal stimuli. Users can perceive and compare two historical temperature recordings by feeling the two stimuli in succession. Rather than conveying absolute temperature values and engages users in a top-down cognitive process, this concept foregrounds the sensory difference, supporting experiential comparison of gradual change over a large temporal and spatial scale.

3.2 System Implementation

The prototype consists of two main interfaces, a web based GUI to allow selection by year, season, and country, and a physical prototype device to deliver thermal feedback.

Physical Prototype and Implementation

The physical prototype contains two Peltier modules (*TEC-12706*) to allow the presentation of two thermal stimuli simultaneously. The Peltier modules are powered and controlled through a dual DC motor controller (*ZK-BM1*). Three infrared temperature sensors (*MLX90614*) are used to monitor the temperature of the Peltier modules and to measure the skin temperature of the user. Additionally an environmental sensor (*BME280*) is used to measure the ambient temperature and humidity. These temperature and environmental sensors ensure the accuracy and safety of the provided thermal stimuli. To clearly indicate the correspondence between the thermal stimuli and the selected data points, two 2.25-inch LCD screens

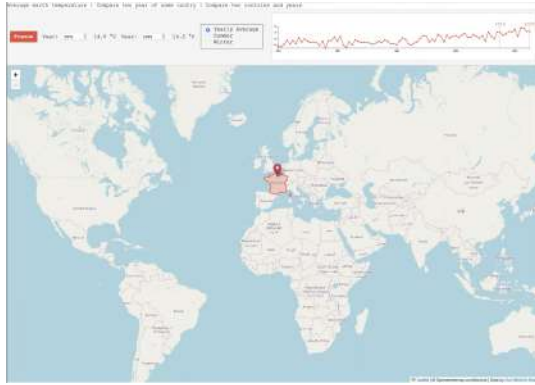


Figure 2: Example interaction. On the GUI (right), the country France is selected on the map, with the years 2015 and 2025 chosen and the temperature record set to yearly average. These selections are used by the physical prototype to control both the display and the corresponding thermal stimuli. Users use their fingertip to feel the thermal stimuli.

are used to display textual information. All the above mentioned components are controlled by a *ESP32* equipped with an *MicroSD* Card Module for memory expansion.

Graphical User Interface

The system includes a web-based graphical user interface (GUI) hosted directly on the *ESP32*, with climate data stored locally on the device. This design allows the prototype to be accessed using any WiFi-enabled device with a standard web browser, without requiring external servers or software installation. Through the interface, users can select climate data points for comparison, such as year, season, or geographic location through graphical widgets. Beyond inputs, the GUI also provides basic visualisation such as line chart of temperature over time and a world map to guide exploration. The temperature record data is sourced from *Our World in Data*¹. The map data comes from *OpenStreetMap*² and is rendered using the open-source library *Leaflet*³.

Mapping Climate Temperature Data to Thermal Sensation

To translate abstract climate temperature values into meaningful and repeatable thermal stimuli, we adopt an established approach from prior work [15]. Specifically, a variant of the Predicted Mean Vote (PMV) model [29] is used to estimate whole-body thermal sensation under a given environmental condition defined by a set of parameters including air temperature. To determine the thermal conditions that the Peltier modules must recreate in order to evoke a comparable thermal sensation at the fingertip, we rely on [36]’s model for predicting local thermal sensation. This approach allows us to map consistently and continuously climate temperature data to thermal stimuli in a repeatable manner, supporting the sensory comparison interaction technique. Further details of the validation and calibration process are provided in [15].

4 Demo Scenarios

For this demonstration, we envision three interaction scenarios structured to progressively guide exploration of climate data through

thermal sensation. Together, these scenarios highlight both the scale and heterogeneity of climate change, supporting experiential comparison of temperature records across time and locations.

The thermal stimuli generated by the physical prototype are strictly controlled in accordance with established guidelines for the use of thermal simulation in interactive systems [34] to guarantee safety. The physical prototype is calibrated to each attendee by adapting the provided thermal stimuli according to their fingertip skin temperature. For optimal experience, one attendee engages directly with the system at a time. However, additional attendees can follow the interaction through the GUI, which displays the selected data points and the exploration process, allowing spectators to understand and discuss the interaction as it unfolds.

4.1 Scenario 1: Sensory overview of global warming

In this scenario, attendees explore long-term trends in global average surface temperature by selecting and comparing temperature records from two different years via the GUI (e.g. by using sliders on the line chart, buttons or text inputs). The physical prototype translates the temperature values into thermal stimuli through the Peltier module. Attendees can then directly compare the felt sensation of the two stimuli to grasp the effects of global warming on a global level. Attendees are also encouraged to vary the temporal distance between the selected data points, this helps attendees to develop an experiential understanding at a global scale of how slow, incremental temperature changes accumulate into perceptible differences over large temporal scales.

4.2 Scenario 2: Sensory exploration of country-level temperature evolution

This scenario offers a focused exploration of localised climate change effects. Attendees can select a specific country by drag and drop the map pin or click on a specific country on the world map. They then explore changes in recorded temperatures (annual, summer, or winter averages) by selecting and comparing two years and compare the sensation through the same procedure as scenario

¹<https://ourworldindata.org/grapher/monthly-average-surface-temperatures-by-year>

²<https://www.openstreetmap.org>

³<https://leafletjs.com>

1. Attendees are encouraged to compare the felt sensation with their personal memories, and extend the exploration deeper in time to support beyond their direct experience of a particular country, to support reflection on how climate change extends beyond direct experience.

4.3 Scenario 3: Sensory comparison of heterogeneous climate change impacts and climate analogue

In the third scenario, attendees explore and compare the heterogeneous impacts of climate change across different countries. Attendees can select two countries on the world map using map pins and compare their temperature records across extended time scales following similar procedures in scenario 1 and 2. They can then compare the changes in temperature over large time scale of both countries through comparing the felt thermal sensation.

The scenario also includes a climate analogue experience, in which users select a country of interest and its projected future climate analogue. By experiencing the thermal difference between the two, attendees can project themselves into future climate conditions and reflect on uneven climate trajectories.

Through this interaction, the system foregrounds the uneven magnitude and trajectory of temperature change in different countries, making it clear that climate change does not affect all countries equally through sensory comparison. Further, by engaging with these contrasts through thermal sensation, the scenario aims to evoke empathy and encourage reflection on spatial disparity, unequal exposure, and the differentiated consequences of global warming.

5 Limitations and Future Work

Our current prototype focuses exclusively on historical climate data. Future work could extend the system to include projected climate scenarios, enabling sensory comparison that supports reflection on future trajectories and climate-related risks. Such extensions would allow the thermal comparison technique to be applied not only to retrospective understanding, but also to anticipation. Future iterations could also incorporate finer-grained, localised climate data, as well as data related to extreme weather events such as heatwaves. Integrating these data sources could further emphasise the localised and often severe impacts of climate change, and help make perceptible its non-linear dynamics and complexity through embodied thermal experience.

6 Conclusion

This paper introduces a novel interaction technique for climate change communication and demonstrates its feasibility through an interactive system. By enabling users to explore and understand climate data through thermal stimuli, our work highlights the potential of bottom-up, sensory-driven interaction to complement existing visual and analytical approaches.

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